

RESEARCH ARTICLE

LLM-Based XAI Framework for Neurofibromatosis Classification

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ABSTRACT Half of individuals with neurofibromatosis (NF) inherit the disorder from an affected parent, while the remaining cases arise from sporadic. Neurofibromatosis data is generally tabular that ML algorithms have shown successful results on tabular data. However, insufficient data is a significant factor affecting performance. There is also another limitations as clinical interpretability—an essential requirement for trust, transparency, and ethical use in medical decision making. Therefore, the proposed frameworks bridges data augmentation, ML interpretability with natural language explanation. Extreme Gradient Boosting (XGBoost) emerges as the best-performing model, achieving approximately 0.80 accuracy and 0.80 F-score, substantially improving upon previously reported results on the same dataset. The integration of Explainable Artificial Intelligence (XAI) techniques such as SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-agnostic Explanations), and PFI (Permutation Feature Importance) becomes crucial for uncovering the underlying relationships between genetic, phenotypic, and demographic features. Furthermore, integrating Large Language Models (LLMs) enables the automatic interpretation and textual summarization of XAI outputs, translating complex importance maps and feature interactions into human-readable clinical insights. These XAI artefacts are translated into concise clinical narratives by the Phi-3-mini LLM, and the resulting explanations are quantitatively assessed with global/local faithfulness, hallucination rate, and feature- and text-level consistency metrics, demonstrating high alignment between narrative and XAI evidence after prompt optimization. Consequently, the proposed framework delivers a transparent and auditable decision-support pipeline for health dataset and offers a reusable template for integrating XAI and LLM-based explanation in rare-disease tabular modeling. This hybrid framework bridges the gap between computational reasoning and medical understanding. This framework give us to knowledge-driven decision support system for NF diagnosis and risk assessment. The proposed approach not only improves model explainability but also promotes trustworthy AI practices in rare genetic disease research.

INDEX TERMS XAI, LLM, machine learning, AI.

I. INTRODUCTION

Neurofibromatosis (NF) is a group of rare genetic disorders that occurs in one of every 3000 or 3500 individuals [1], [2], [3]. Around the half of NF cases arise sporadically from de novo germline mutations without a familial background. The disease exhibits wide clinical variability, including café-au-lait macules, cutaneous and plexiform neurofibromas, freckling, and malignant peripheral nerve

sheath tumors. Despite its autosomal-dominant inheritance, NF1 shows unpredictable symptom expression and lacks clear genotype–phenotype correlations, likely influenced by modifier genes and environmental factors such as nutrition and lifestyle [3].

In recent years, the rapid advancement of Machine Learning (ML) and Artificial Intelligence (AI) has resulted in a proliferation of models designed to support decision-making across diverse domains, including healthcare, finance, and criminal justice. While the pursuit of higher predictive accuracy has driven significant progress, it has also

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introduced critical challenges regarding data availability, model transparency and interpretability. Firstly, tabular data has limited availability of patient data, class imbalance and high heterogeneity. Developing accurate and interpretable diagnostic models remains a challenging task in medical data. For example, the class imbalance mainly effects the model to favor the majority class, reducing its ability to correctly identify minority samples. Despite their limited size, small tabular datasets can reveal substantial analytical value and has shown better performance with using ML when suitable data augmentation strategies are employed. Tabular data holds great potential for knowledge discovery and informed decision-making. Synthetic data generation has relied on traditional augmentation methods such as Gaussian noise, Bayesian modeling and randomization-based approaches to enrich these datasets [4]. Secondly, with using broad sense of AI, closed box systems, whose inner workings are often opaque to end users and domain experts, are used in medical data analysis that lead to non understandable models. This closed box prediction directly influence human lives and societal outcomes that poses serious ethical and practical risks. In view of the this concern, the European Union's General Data Protection Regulation (GDPR) have emphasized the "right to explanation," underscoring the need for models whose reasoning processes can be understood and justified. Consequently, Explainable Artificial Intelligence (XAI) has emerged, aiming to develop methods that make AI systems more transparent, interpretable, and accountable to human understanding [5].

Building upon this motivation, the present study explores the integration of XAI techniques within the context of NF data analysis. By combining data augmentation strategies with advanced classification models, this research seeks to overcome the challenges posed by limited and imbalanced medical datasets. Furthermore, Large Language Models (LLMs) are employed to interpret and articulate XAI outputs in natural language, transforming complex analytical explanations into accessible clinical narratives. This integrated LLM-XAI framework aims to enhance both the interpretability and usability of AI-driven diagnostic systems for rare genetic disorders such as NF.

Recent literature shows that artificial intelligence for rare diseases has advanced considerably, yet its clinical translation remains constrained by limited cohort sizes, class imbalance, heterogeneous phenotypes, and weak external generalizability, especially in structured clinical datasets [6], [7]. In parallel, explainable AI research in healthcare has moved beyond accuracy-only reporting and increasingly emphasizes the validity, robustness, and clinical usefulness of explanations [8], [9]. Furthermore, recent studies on medical large language models indicate that natural-language outputs should not be judged only by fluency, but also by groundedness, faithfulness, hallucination risk, and alignment with source evidence [10], [11], [12]. These findings collectively support the need for an integrated framework that combines balanced tabular learning, rigorous XAI analysis,

and source-grounded narrative explanation for rare-disease clinical decision support.

Due to the limited availability and heterogeneous nature of NF datasets, the proposed framework employs data augmentation techniques to expand the sample size and improve the diversity of training instances. It is also adopt class distribution equalization strategies to mitigate the effects of class imbalance that ensure the fair model learning. Subsequently, several ML algorithms are developed and evaluated to predict NF-related clinical outcomes. The best predictive model is selected to examine using XAI techniques, including SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-agnostic Explanations), and PFI (Permutation Importance Analysis). Finally, the XAI outputs are interpreted by a LLM which gives to technical explanations into doctors and patients. This framework enhances both the interpretability and clinical applicability of the predictive models, bridging the gap between computational intelligence and medical decision support. The main contributions of this study an be summarized as follows:

- A customized data augmentation framework is designed for addressing the scarcity of NF data. By designing this framework, statistically syhnetic samples generated to enhance model robustness. Advanced resampling and balancing techniques are applied to ensure fair class representation, reducing bias toward the majority class and improving the recognition of minority outcomes.
- The proposed framework proposes an integrated framework that bridges ML interpretability with natural language explanation, promoting transparency, reproducibility, and high trustworthiness in AI-based healthcare systems.
- Multiple classification algorithms are employed and systematically compared to identify the most accurate and generalizable model for NF outcome prediction.
- The best-performing model is interpreted using by XAI methods which uncover the feature level reasoning behind model predictions and provide transparency in clinical decision support. A domain-adapted LLM is employed to analyze and articulate XAI results, transforming technical explanations into coherent, human-readable medical insights. The LLM results are evaulated with lobal/local faithfulness, hallucination rate, and feature- and text-level consistency metrics.

This study is organized as follows. Section I provides a comprehensive overview of prior work related to ML and XAI methods applied to Neurofibromatosis dataset. Section II introduces the proposed framework. Section III gives details of the experimental results. Section IV presents and discusses the results of the empirical analyses, highlighting the comparative performance of classical models and LLM-based variants as well as the contribution of XAI methods to clinical interpretability. Finally, Section V concludes the paper by summarizing the key findings, emphasizing the advantages of the proposed framework, and outlining potential directions for future research.

II. RELATED WORKS

In this section, It is examined the representative studies employing ML methods for Neurofibromatosis Type 1 (NF1) diagnosis (Table 1). *NF1* has gained increasing attention across clinical, genomic, and imaging domains with the improvement of ML and AI. Due to the rarity and heterogeneity of NF1, most existing studies face challenges related to limited data availability, class imbalance, and lack of interpretability.

Morris et al. [1] performs predictive modelling on NF1 electronic health record (EHR) samples to identify comorbidities such as optic pathway glioma and ADHD, achieving AUROC values between 0.74 and 0.82 using gradient boosting. Sharafi et al. [3] introduced a tabular dataset containing 241 NF1 patients (121 sporadic, 120 familial) to predict NF1 disease. The authors evaluate the models including KNN, ANN, SVM, Decision Tree, and XGBoost. The study achieved a maximum accuracy of 62.86% using XGBoost. However, this study has not employed any XAI framework, leaving the reasoning behind predictions opaque. While both studies are validated the feasibility of structured data learning, but, interpretability analysis is not held in papers. These studies confirm the predictive value of structured clinical variables for NF1-related tasks. However, they place limited emphasis on explanation quality and clinical interpretability, which remain essential for reliable rare-disease decision support.

Pinto et al. [17] used artificial neural networks and statistical learning methods to evaluate the structural impact of NF1-related protein mutations. The authors show that data-driven models can infer mutation severity from protein sequence and biochemical features. Lakshmi et al. [19] extended this paradigm by building a prognostic ML framework for generic genetic disorder prediction, underlining the potential for such models to generalize beyond specific diseases like NF1. Tomkinson et al. [21] investigated NF1 genotype related patterns using high-content microscopy and machine learning on Schwann cells. The study indicates that AI can capture disease-relevant cellular morphology signatures at the microscopic level. Nevertheless, its data modality and experimental setting differ substantially from small structured clinical NF1 datasets used for tabular prediction. These studies show that AI can detect important biological patterns. However, they use different goals and data types than small structured NF1 datasets, so the need for explainable tabular classification still remains.

Zhang et al. [13] proposed Deep Interactive Networks (DINs) for automatic segmentation of plexiform neurofibromas in whole-body MRI, reporting a 44% improvement in Dice coefficient relative to conventional CNNs. Kolokolnikov et al. [14] combined 3D U-Net architectures with radiomics features to achieve a 68% per-scan Dice improvement across multi-center datasets. The authors designs three stages as anatomy segmentation, NF segmentation, and tumor candidate classification. Wei et al. [15]

introduced a multicenter deep learning framework for early automated screening of malignant peripheral nerve sheath tumours (MPNSTs), obtaining about 85% classification accuracy on 347 patients. Despite impressive results, most imaging pipelines remain “closed box” models, lacking formal XAI mechanisms that connect pixel-level outputs with clinical semantics. Image-based models have achieved promising results in NF1 analysis. However, their dependence on modality-specific data and complex pipelines limits their direct applicability to structured clinical datasets. Beyond NF1, parallel advances in other genetic and neurodegenerative diseases provide methodological inspiration. Velez et al. [23] proposed a comprehensive ML framework for predicting the age of onset in familial and sporadic Alzheimer’s disease using gradient boosting and ensemble learning. Although the biological system differs, this study offers a close analogy to familial/sporadic differentiation in NF1 and demonstrates the importance of balanced datasets and model calibration for early-onset prediction. Genetic ML efforts have also been reviewed by Faria et al. [18], who conducted a scoping review on the application of ML to genetic diagnosis tasks. They highlighted three persistent issues: limited dataset curation, lack of standardized interpretability metrics, and underrepresentation of tabular genotype–phenotype relationships in published work. Kaster et al. [20] compared rule-based and large language model-based phenotype extraction from NF1 clinical notes. Their study shows that recent NF1 research is also expanding toward language-based analysis of unstructured medical records. However, the task is centered on phenotype extraction from text, rather than explaining predictions from structured tabular models through XAI-guided clinical narration. Grech et al. [16] conducted a systematic review focusing on NF1 variant detection pipelines, concluding that transparency, dataset diversity, and integration of interpretability remain major limitations across rare-disease ML literature. Hellmann et al. [22] reviewed artificial intelligence-based tools for precision diagnosis and treatment in NF1-associated peripheral and central glial tumors. Their work highlights the growing role of AI across NF1-related clinical decision pathways and supports the broader movement toward more precise and data-driven analysis. At the same time, it also underlines that NF1 research remains methodologically diverse, which increases the need for transparent and clinically interpretable AI frameworks. Overall, the literature demonstrates substantial progress in the predictive and diagnostic capacity of ML for NF1 predictions. Nevertheless, recurring limitations persist: (1) data scarcity and imbalance; (2) underrepresentation of tabular and mixed-modality datasets; (3) minimal adoption of formal explainability frameworks; and (4) limited translation of computational results into human-interpretable narratives. To bridge these gaps, this study focuses on NF1 tabular clinical data augmented through synthetic oversampling and balancing, applies multiple ML algorithms to identify

TABLE 1. Comparison of representative ML studies on NF and related genetic/clinical disorders.

#	Study	Dataset (n)	Method	Performance Metric	XAI	Tabular Data
1	[3]	241 (121 sporadic / 120 familial)	KNN, ANN, SVM, DT, XGBoost	Accuracy = 62.86% (XGBoost)	No	Yes
2	[1]	798 EHR records	Gradient Boosting	AUROC = 0.74–0.82	No	Yes
3	[13]	3 MRI datasets	Deep Interactive CNN	Dice +44% vs baseline	No	No
4	[14]	Multi-center WB-MRI	3D U-Net + Radiomics	Dice +68% per scan	No	No
5	[15]	347 subjects (7 centres)	Context-aware DL classifier	Accuracy $\approx$ 85% (test)	No	No
6	[16]	Multiple datasets (review)	Comparative / Survey	–	–	–
7	[17]	Proteins/mutations dataset (NF1 context)	ANN + ML techniques	Reported improved prediction of mutation impact [17]	No explicit XAI	Yes (protein/mutation tabular features)
8	[18]	Genetic-diagnosis studies (review)	Survey of ML use in genetics	N/A	Yes (discusses XAI)	Mixed
9	[19]	Emerging genetic-disorder dataset	ML prognostic modelling	Early results reported	No explicit XAI	Yes (tabular genetic)
10	[20]	NF1 clinical notes	Rule-based NLP vs LLMs	F1-based evaluation; LLM phase-to-phase drop = 4.4–5.1%, rule-based = 8.8%	No	No
11	[21]	20,680 Schwann cells	Cell Painting + Logistic Regression	Balanced Accuracy = 0.80 (test)	No	No
12	[22]	Public NF1 AI datasets (review)	Review / dataset survey	N/A	Mixed	Mixed
13	This Study	NF1	ML and LLM prediction model	Accuracy, F-score, Precision, and Recall	XAI + LLM explain	Yes (tabular)

the best predictor. SHAP, LIME, and PFI are employed for interpretability, and finally leverages a domain-specific Large Language Model (LLM) to translate XAI outputs into clinician-readable explanatory text. This integration of data augmentation, interpretability, and LLM-based explanation represents a novel contribution to explainable precision modelling for rare genetic diseases.

III. PROPOSED FRAMEWORK

In prediction applications, ML algorithms enables us to enhance the prediction performance of given datasets. However, limited availability and heterogeneous nature of health datasets brings difficulty in predicting performance and makes it difficult to come up with real life applications. Understanding closed box system and converts to explanation to domain experts is another important problem that have to be solved. In this study, it is proposed a framework that predict NF disease in health domain with the AI integration.

To present the proposed method more clearly, this section introduces the framework as an end-to-end workflow that starts from raw NF1 tabular data and ends with both predictive and explanatory outputs. The pipeline first prepares the data through cleaning, preprocessing, class balancing, and augmentation in order to address the limitations of small and heterogeneous clinical datasets. Then, It evaluates multiple candidate predictors under a unified cross-validation protocol and selects the best-performing model according to classification performance. After model selection, the framework applies complementary XAI methods to generate global and local explanations of the final decision process. In the last stage, these XAI artefacts are converted into clinician-readable narratives by an instruction-tuned LLM. The generated explanations are further assessed with faithfulness, hallucination, and consistency measures. In this way, the proposed framework combines prediction, interpretability, and narrative explanation within a single auditable pipeline.

A. PROPOSED FRAMEWORK ARCHITECTURE

I provide a clear overview of the proposed analysis pipeline, Fig. 1 illustrates the overall workflow of the NF1 XAI-LLM framework. The framework begins with tabular clinical data preparation and proceeds through preprocessing, balancing, and model evaluation stages. The best-performing model is then analyzed using multiple explainable AI methods, and the resulting evidence is transformed into clinician-readable explanations through an instruction-tuned language model. Finally, the generated explanations are evaluated using several consistency and faithfulness metrics. The proposed framework consist of 9 steps which is shown in Fig. 1. To explain in detail, an end-to-end classification, XAI and LLM based explainable pipeline is designed for tabular datasets that are both small in size and affected by class imbalance. In the first step, the data is processed for prediction step. In the subsequent feature-cleaning step, columns with a single unique value, those with a high proportion of missing observations, and those that are highly correlated with each other is removed, thereby reducing noise and mitigating multicollinearity effects. During preprocessing, missing values are imputed with the median, and all numerical features are standardized to prevent modeling biases arising from heterogeneous feature scales. The class imbalance in the dataset is addressed by first applying SMOTE to enrich minority classes with synthetic samples. Subsequently, it is performed data augmentation via Gaussian noise injection and bootstrap resampling. This provide us that enlarged the effective training set and improved the model's ability to generalize from limited data. In the modeling step, five algorithms are employed as logistic regression, random forest, XGBoost, LLM baseline and LLM LoRA which are defined and evaluated under five-fold stratified cross-validation using accuracy, precision, recall, and F-score as performance criteria. Furthermore, the model achieving the highest F-score is retrained on the entire balanced and augmented dataset and selected as the final classifier. To enhance the interpretability of this model's decision process, global, sample-based SHAP importances are first computed to identify which features played a dominant role in predictions. Then, LIME explanations are generated for a single instance to expose the model's local decision logic, and PFI is conducted to verify the consistency of the SHAP-based findings. Finally, the LLM based narrative generation step is incorporated that the model performance summary together with the SHAP, LIME, and PFI methods. They are packaged into a structured prompt and fed to an instruction tuned language model, which automatically produced a coherent, human-readable explanation of the pipeline's behavior. This last component enables data scientists and domain experts to consume the XAI results faster, in a unified form, and without manually interpreting multiple separate plots. Figure 1 summarizes the full analytical workflow of the proposed method. The framework is organized into four connected stages: data preparation, predictive modeling,

explainability analysis, and narrative synthesis. In the first stage, the NF1 dataset is cleaned, imputed, standardized, balanced, and augmented to produce a more robust training set. In the second stage, candidate models are trained and compared under stratified five-fold cross-validation, and the highest-performing model is retained as the final predictor. In the third stage, SHAP, LIME, and PFI are applied to this selected model to obtain complementary global and local explanations. In the final stage, the XAI outputs and model performance summaries are transformed into human-readable clinical narratives by the Phi-3-mini model, and the generated narratives are quantitatively evaluated for alignment with the underlying XAI evidence.

The outputs of the framework are produced progressively across these stages. The modeling stage yields comparative classification scores for all candidate predictors, from which the best model is selected. The explainability stage produces feature-level and instance-level interpretation outputs in the form of SHAP, LIME, and PFI analyses. The final LLM stage converts these analytical artefacts into concise natural-language summaries and evaluates them using global/local faithfulness, hallucination rate, and consistency metrics. Therefore, the framework does not stop at prediction accuracy alone, it also provides traceable evidence about why the final model performs as it does and how its behavior can be communicated to domain experts.

B. XAI METHODS

1) SHAP (SHAPLEY ADDITIVE EXPLANATIONS)

SHAP is an explanation framework that interprets a model prediction. In detail, It give as the outcome of a cooperative game in which features act as players, each receiving a Shapley value equal to its average marginal contribution over all possible coalitions of features. This guarantees desirable properties such as local accuracy and consistency, which makes SHAP suitable for understand complex tabular models [24], [25]. SHAP has been employed various application to explain the closed box models that give us local or global outcomes [26]. SHAP quantifies how much each individual feature contributes to a model's output in both ML and deep learning algorithms. Unlike conventional importance measures, it can disentangle these contributions per class, which makes it applicable even to multi-class classification problems. By computing SHAP values, the model's prediction is decomposed into feature-level effects, allowing practitioners to see which inputs pushed the prediction toward a certain class and by what magnitude. In this way, SHAP substantially enhances the interpretability and transparency of otherwise opaque ML and DL models [27].

2) LIME (LOCAL INTERPRETABLE MODEL-AGNOSTIC EXPLANATIONS)

LIME provides a model-agnostic strategy that focuses on explaining one prediction at locally [26]. It perturbs the

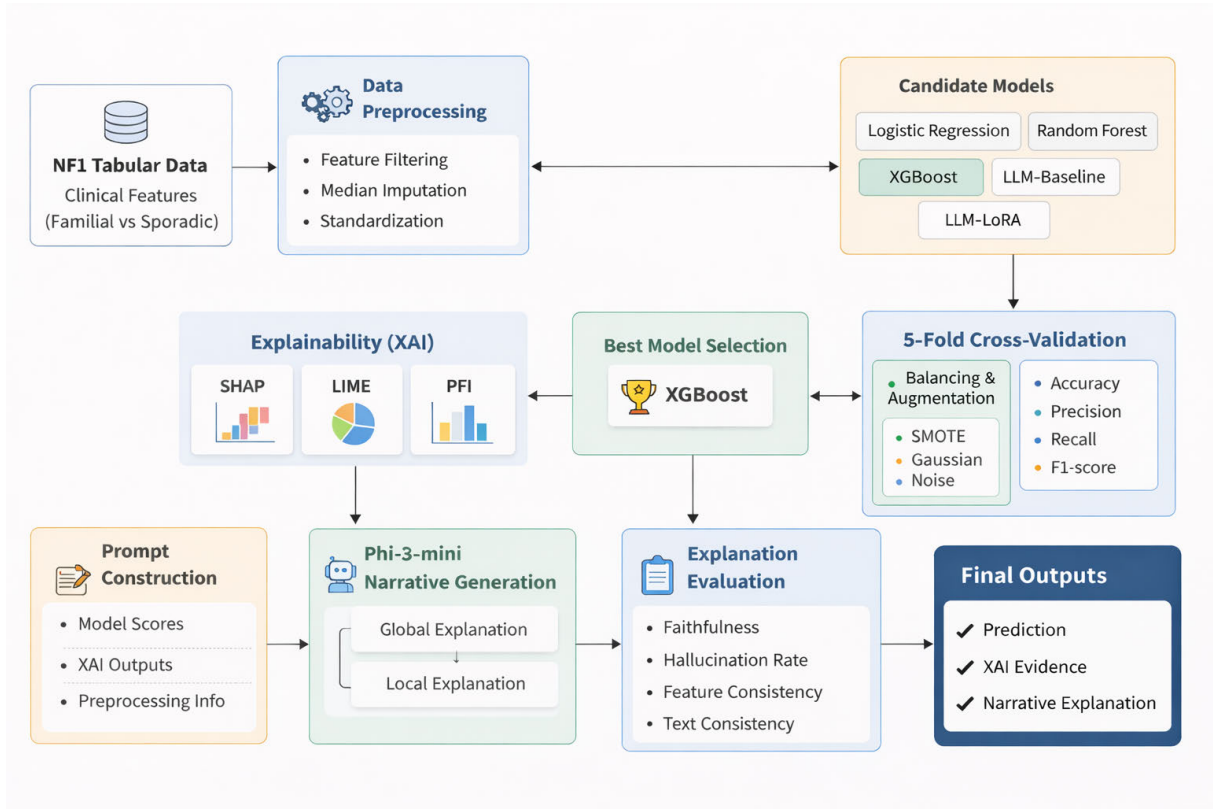


FIGURE 1. Proposed model framework.

original instance to generate synthetic neighbors. Furthermore It queries the closed box model for each neighbor, and then fits a sparse, locally weighted linear model that mimics the closed box behavior in that small neighborhood [28]. Because the surrogate is only required to be faithful locally, LIME can turn highly nonlinear or ensemble decision boundaries into a human-readable list of feature contributions [29], [30]. At the same time, the literature notes that LIME's explanations may vary with the perturbation strategy or neighborhood definition, so combining LIME with more axiomatic methods such as SHAP often yields more robust interpretability in practice [28].

3) PERMUTATION FEATURE IMPORTANCE (PFI)

PFI model is agnostic method that processed to identify the top features as most critical with take care of model accuracy and efficiency [31]. Permutation feature importance quantifies the relevance of a feature by measuring the deterioration in predictive performance when the feature's values are randomly permuted across samples [32], [33].

C. CLASSIFICATION ALGORITHMS IN THE PROPOSED FRAMEWORK

1) LOGISTIC REGRESSION

Logistic regression is a classical model that estimates the conditional probability of a categorical outcome.

It is a long-established statistical method that has regained prominence over the last few decades for modeling categorical outcomes. It is particularly well suited to binary classification problems, where the response can take only two possible values, typically coded as 0 and 1. References [34], [35]. LR maps a linear combination of input features through a logistic (sigmoid) function. The parameters are typically learned via maximum likelihood, and ℓ_2 or ℓ_1 regularization is introduced to prevent overfitting, which is particularly important in small-sample, high-dimensional tabular problems [36], [37].

2) RANDOM FOREST

Random forest is an ensemble method that combines a large number of decision trees grown on bootstrap-resampled subsets of the training data [38]. It can be defined as a decision tree-based approach and randomly constructs multiple decision trees [39]. The final prediction is obtained by majority voting or averaging, yielding a model that is robust to noise, nonlinearities, and moderately missing data [40]. Owing to its bagging-based construction and feature subsampling, random forest is well suited to heterogeneous tabular datasets and often provides competitive performance with minimal tuning [41].

3) XGBOOST

XGBoost is a gradient-boosted decision tree framework that builds an additive ensemble in a stage-wise manner, where

each new tree is trained to fit the negative gradients (residuals) of the current model [42]. Its objective integrates both training loss and a structural regularization term that penalizes overly complex trees, leading to improved generalization on small or noisy datasets. System-level optimizations such as histogram-based split finding, column subsampling, sparsity awareness, and parallelization which make XGBoost highly efficient and a de facto standard for tabular data in real world classification tasks [43].

D. LLM LAYER

LLMs are Transformer-based architectures trained on massive text corpora with self-supervised objectives which is typically next-token prediction. Therefore, they acquire broad linguistic competence and emergent in-context learning abilities [44], [45]. After this pretraining phase, instruction-tuning and alignment techniques are applied to make the models follow natural-language instructions and produce helpful, safe outputs for downstream tasks [46]. In model-interpretation pipelines, such LLMs can be used not to generate the primary prediction, but to *verbalize* the outputs of other models (e.g., SHAP tables, LIME local rules, or permutation-importance scores), turning technically dense artifacts into coherent, user facing narratives. Within this work, the explanation layer is implemented with the `microsoft/Phi-3-mini-4k-instruct` model, a compact, instruction-tuned member of the Phi-3 family released by Microsoft. This model is trained on carefully filtered, high-quality data and aligned to respond to structured prompts, which allows it to ingest heterogeneous analytical artifacts and render them into fluent English reports suitable for data scientists and domain experts. In other words, it acts as a narrative wrapper on top of the XAI components, improving the communicability and auditability of the entire pipeline [47].

IV. EXPERIMENTAL EVALUATION AND RESULTS

A. DATASET

In this framework, It is selected the dataset of NF1 from UCI ML Repository. The data consist of routinely observed clinical signs such as cutaneous findings, tumor-related manifestations, and other NF1-associated complications. It is used national NF1 database with the explicit aim of enabling ML approaches to discriminate between familial and sporadic. Additionally, the dataset is tabular, contains 19 features, and is explicitly marked for a classification task [3]. The dataset contains 241 patient records in total, comprising 121 sporadic cases and 120 familial cases, representing a near-balanced class distribution. A stratified 5-fold cross-validation protocol is adopted, where each fold uses approximately 193 samples for training and 48 samples for validation. Data augmentation is applied exclusively within each training fold to prevent data leakage. The description of the NF1 dataset is shown in Table 2. After computing the pairwise Pearson correlation matrix across all 19 features, no feature pair exceed the

TABLE 2. Variables of the NF1 clinical symptoms dataset.

Variable Name	Role	Type	Missing Values
With Tumors	Feature	Binary	no
Healthy	Feature	Binary	no
Age of Mother	Feature	Integer	yes
Age of Father	Feature	Integer	yes
Age at First Diagnosis	Feature	Integer	yes
Café au lait (CLS)	Feature	Binary	no
Axillary Freckles	Feature	Binary	no
Inguinal Freckles	Feature	Binary	no
Lisch Nodules	Feature	Binary	no
Dermal Neurofibromins	Feature	Binary	no
Plexiform Neurofibromins	Feature	Binary	no
Optic Glioma	Feature	Binary	no
Skeletal Dysplasia	Feature	Binary	no
Learning Disability	Feature	Binary	no
Hypertension	Feature	Binary	no
Astrocytoma	Feature	Binary	no
Hamartoma	Feature	Binary	no
Scoliosis	Feature	Binary	no
Other Symptoms	Feature	Binary	no
Case Type	Target	Binary	no

0.9 correlation threshold. Therefore, no feature is removed during the correlation-based cleaning step. The heatmap of the dataset is given in Figure 2.

B. PERFORMANCE METRICS

In this study, the classification step is assessed using four performance metrics as accuracy, precision, recall, and F_1 -score. The dataset contain N samples and let $\mathcal{C} = \{1, 2, \dots, K\}$ denote the set of target classes, where in our case the clinically relevant target is

$$\text{Case Type} \in \{\text{sporadic, familial}\}.$$

For each class $c \in \mathcal{C}$, I define the usual counts: true positives TP_c , false positives FP_c , false negatives FN_c , and true negatives TN_c .

The accuracy measures the proportion of correctly classified instances over all instances.

$$\begin{aligned} \text{Accuracy} &= \frac{\text{number of correct predictions}}{\text{total number of samples}} \\ &= \frac{1}{N} \sum_{i=1}^N \mathbf{1}(\hat{y}_i = y_i), \end{aligned} \quad (1)$$

where y_i is the ground-truth label and $\hat{y}_i$ is the predicted label for sample i .

The code evaluates models on a imbalanced clinical dataset, it uses precision so that each class contributes equally, independent of its frequency.

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad (2)$$

Recall captures the model's ability to recover positive cases in each class, which is important in medical/clinical screening scenarios.

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}, \quad (3)$$

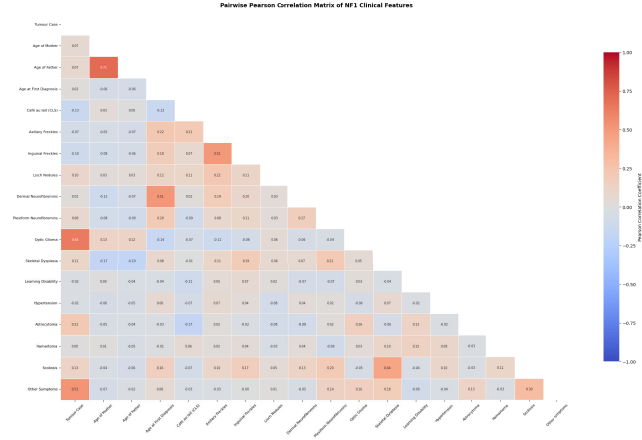


FIGURE 2. Pairwise pearson correlation matrix of NF1 clinical features.

To combine precision and recall into a single, class-balanced indicator, the pipeline reports the F -score. For each class c :

$$F_{1,c} = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}, \quad (4)$$

In the LLM Based Narrative Generation step, faithfulness, hallucination rate and consistency metrics are employed to evaluate the LLM's output. Before presenting the formal definitions, I briefly summarize the intuition behind each metric.

Global and local faithfulness answer the question: “Did the narrative mention the features that XAI identified as important?”

Hallucination rate answers: “Did the narrative mention any features that were not supported by any XAI method?”

Feature-level consistency measures whether two narratives generated for the same instance highlight the same set of features.

Textual consistency measures whether the overall meaning of two narratives is semantically similar. Together, these four metrics provide a quantitative audit of how faithfully and stably the LLM translates XAI evidence into natural language.

In the LLM Based Narrative Generation step, faithfulness, hallucination rate and consistency metrics are employed to evaluate the LLM's output.

$$\text{Faithfulness}_{\text{global}}(N) = \frac{|F^{\text{global}} \cap F^{\text{narr}}(N)|}{|F^{\text{global}}|}, \quad (5)$$

$$\text{Faithfulness}_{\text{local}}(N) = \frac{|F^{\text{local}} \cap F^{\text{narr}}(N)|}{|F^{\text{local}}|}. \quad (6)$$

$$\text{HallucinationRate}(N) = 1 - \frac{|(F^{\text{global}} \cup F^{\text{local}}) \cap F^{\text{narr}}(N)|}{|F^{\text{narr}}(N)|}. \quad (7)$$

$$\text{Consistency}_{\text{features}}(N_1, N_2) = \frac{|F^{\text{narr}}(N_1) \cap F^{\text{narr}}(N_2)|}{|F^{\text{narr}}(N_1) \cup F^{\text{narr}}(N_2)|}. \quad (8)$$

$$\begin{aligned} \text{Consistency}_{\text{text}}(N_1, N_2) &= \cos(e(N_1), e(N_2)) \\ &= \frac{e(N_1)^\top e(N_2)}{\|e(N_1)\|_2 \|e(N_2)\|_2}. \end{aligned} \quad (9)$$

Notation. Here, F^{global} denotes the set of globally important features (e.g., top- k features according to SHAP and permutation importance), F^{local} is the set of locally important features for a given instance (e.g., top- k LIME features), and $F^{\text{narr}}(N)$ is the set of feature names explicitly mentioned in narrative N . The function $e(N)$ denotes a vector representation (embedding) of narrative N , obtained for example via TF-IDF or a sentence-embedding model.

These metrics capture complementary aspects of how well LLM generated narratives align with the underlying XAI evidence. Global and local faithfulness measure the fraction of XAI identified important features (from SHAP, PFI or LIME) that are explicitly reflected in the narrative, following recent work that evaluates the faithfulness of LLM self explanations to their internal reasoning processes [48]. The hallucination rate counts the proportion of mentioned features that are not supported by any of the XAI methods, in line with hallucination benchmarks for large language models that treat unsupported content as hallucinated generations [49]. Feature level consistency uses the Jaccard index to quantify how stable the set of highlighted features is across narratives or runs. It is similar to LLM explainability frameworks such as SMILE that analyse the stability of explanation outputs across instances and samples [50]. Finally, we compute textual consistency as the cosine similarity between the embedding vectors of two narratives, which reflects how close their meanings are. This is consistent with LLM evaluation setups that use cosine similarity between embedded responses to measure semantic alignment and output quality for different chatbots or prompting strategies [51].

C. HYPERPARAMETERS

In this framework, numerous experiments are processed to get best performance for prediction of NF1 dataset. It is adopted grid search technique to get best hyperparameters. The hyperparameters used in the experiments are summarized in Table 3. For the classical models, It is retained a reasonably conservative configuration. Firstly, XGBoost is run with 400 estimators, shallow trees (depth 4), and standard subsampling/column subsampling to control overfitting. Secondly, Random Forest uses 300 trees with depth 8 and class balancing. Thirdly, Logistic Regression is trained with a balanced class weight and 200 iterations. For the LLM baselines, It is fixed the generation settings (no sampling, 32 tokens) to obtain stable class outputs, and for the LoRA variant It is applied a lightweight adapter (rank 16, $\alpha = 32$, dropout 0.05) together with a small batch, few epoch training schedule suitable for Colab scale hardware.

D. PERFORMANCE RESULT OF THE MODELS

Several traditional ML models such as logistic regression, random forest, XGBoost are employed under a 5-fold

TABLE 3. Hyperparameters used for classical models and LLM-based approaches in the NF1 experiments.

Model	Component	Hyperparameters
XGBoost	Classifier	<code>n_estimators = 400</code> <code>learning_rate = 0.05</code> <code>max_depth = 4</code> <code>subsample = 0.8</code> <code>colsample_bytree = 0.8</code> <code>gamma = 0.1</code> <code>tree_method = "hist"</code> <code>eval_metric = "logloss"</code> <code>random_state = 42</code>
Logistic Regression	Classifier	<code>max_iter = 200</code> <code>class_weight = "balanced"</code> <code>solver = "lbfgs"</code>
Random Forest	Classifier	<code>n_estimators = 300</code> <code>max_depth = 8</code> <code>class_weight = "balanced"</code> <code>random_state = 42</code>
LLM Baseline	Generation / Inference	<code>model_name = "microsoft/Phi-3-mini-4k-instruct"</code> <code>max_new_tokens = 32</code> <code>do_sample = False</code> <code>temperature = 0.0</code> <code>pad_token_id = tokenizer.eos_token_id</code> <code>device_map = "auto"</code>
LLM + LoRA	LoRA config	<code>r = 16</code> <code>lora_alpha = 32</code> <code>target_modules = ["q_proj", "k_proj", "v_proj", "o_proj"]</code> <code>lora_dropout = 0.05</code> <code>bias = "none"</code> <code>task_type = "CAUSAL_LM"</code>
	TrainingArguments	<code>per_device_train_batch_size = 2</code> <code>per_device_eval_batch_size = 2</code> <code>gradient_accumulation_steps = 4</code> <code>num_train_epochs = 2</code> <code>learning_rate = 2e-4</code> <code>fp16 = if cuda</code> <code>logging_steps = 10</code>

stratified cross validation setting. The 5-fold stratified cross-validation protocol is adopted, in which each fold uses approximately 80% of the samples for training and 20% for testing. Then, It is reported the averages of the above metrics. Subsequently, the same metrics are reused to evaluate LLM baselines and the LoRA fine tuned LLM. Using the identical metric set for both classical and LLM based approaches ensures that any observed improvement can be attributed to the modeling choice rather than to a change in the evaluation criterion.

Table 4 summarizes the comparative performance of all evaluated models on the NF1 clinical classification task. The experimental results indicate that tree based ensemble learning algorithms show the most effective results for this small, initially imbalanced clinical dataset. Among the classical models, XGBoost achieves the highest scores across all reported metrics as accuracy, precision, recall, and F- score, reaching an accuracy of approximately 0.80 and F- score of about 0.80. The close alignment of these

values indicates that XGBoost is not only accurate but also well balanced across classes, which is particularly desirable in clinical settings where the minority class should not be sacrificed for overall accuracy. The result suggests that the feature space contains non linear and possibly interaction effects that an additive ensemble of trees can exploit effectively. Random Forest and Logistic Regression shows worser performance than XGBoost, respectively (≈ 0.70 F-score) and (≈ 0.55 F-score). This performance gap is meaningful because all models are trained on the same augmented feature space, so the superiority of XGBoost can be attributed to its ability to model nonlinear feature interactions and it handle residual imbalance through its built in weighting and tree growth strategy.

The Random Forest model achieved the second best result, with an accuracy of 0.711 and a $F - score$ of 0.705. Although this performance is clearly below that of XGBoost, it remains a reasonable baseline for tabular clinical data. The performance gap between XGBoost and Random

Forest is consistent with prior evidence that boosting, by reweighting hard to classify instances, can make better use of an augmented dataset than bagging based ensembles. In contrast, Logistic Regression showed a markedly lower performance level (accuracy 0.555, $F - score$ 0.551), which indicates that a linear decision boundary is insufficient to separate familial and sporadic cases in this feature space. This finding reinforces the view that the NF1 variables are organized in a way that requires non linear partitioning.

The two LLM based approaches did not reach the performance of the classical tabular models. The prompt based LLM baseline attained only 0.446 accuracy and 0.403 F_1 , showing that a general purpose language model, used without deep task adaptation, does not acquire a strong enough inductive bias for this structured clinical problem. After applying a lightweight LoRA adaptation, the LLM reached 0.544 accuracy and improved its recall to 0.500, but its precision dropped to 0.272 and the F_1 remained low (0.352). This pattern suggests that LoRA makes the model more sensitive. Besides, it captures more positives. But It does not sufficiently constrain its predictions, leading to many false positives.

In summary, the results show that (i) augmentation made it possible to train high capacity models without immediate overfitting, (ii) XGBoost is the most suitable learner for this problem setting, (iii) linear models underfit, and (iv) generic LLMs, used as classifiers, do not yet replace optimized tabular ML on this NF1 dataset. The combined use of SHAP, LIME, and PFI provides an audit trail that is appropriate for clinical or biomedical reporting and supports the claim that the best model is not only more accurate but also more explainable under the chosen protocol.

To statistically validate the superiority of the proposed model, the Friedman test and pairwise Wilcoxon signed-rank tests are conducted on 10-fold cross-validation F_1 scores. The Friedman test confirmed a highly significant difference among the three classifiers ($\chi^2 = 20.00$, $p < 0.001$). Wilcoxon signed-rank tests further demonstrated that XGBoost significantly outperformed both Logistic Regression and Random Forest across all folds ($W = 0.000$, $p = 0.002$), where $W = 0.000$ indicates consistent superiority in every single fold without exception. To further validate that the proposed model reached its best attainable solution, convergence and stability analyses are conducted. It is demonstrated a consistent and monotonic reduction in validation log-loss, plateauing toward the later boosting rounds without divergence or performance degradation, confirming that the model reached a stable optimum within the constraints of the dataset size. Convergence stability is further assessed over 30 independent random seeds, where the optimal iteration showed minimal variance ($\bar{x} = 997.3 \pm 2.3$) and the macro F_1 score remained consistent across all runs ($\bar{x} = 0.671 \pm 0.033$), providing statistical evidence that the reported accuracy of 0.804 is reproducible and not attributable to random initialization.

E. XAI RESULTS

To further validate the reliability of the classification pipeline, It is applied a multi perspective XAI analysis to the best performing model which is the XGBoost classifier achieves the highest accuracy and in experiments. The explainability layer (SHAP, LIME, and PFI) also fits well with these findings. Because XGBoost is the best model, global SHAP values can be interpreted as a ranked list of clinically informative variables, while LIME provides local, per patient rationales.

Below It is discussed the three complementary views obtained from LIME, PFI, and SHAP; the referenced figures correspond to the attached visualizations.

The SHAP summary in Figure 3 gives the most fine grained explanation. It shows, for every feature, the distribution of SHAP values across all samples, with color encoding the underlying feature value. It is inferred three observations stand out:

- 1) **Age related features dominate.** *Age at First Diagnosis*, *Age of Mother*, and *Age of Father* occupy the top rows, which is consistent with the PFI ranking (Fig. 5). This cross method agreement increases confidence that these features are genuinely informative rather than artefacts of SMOTE, Gaussian noise, or bootstrap augmentation.
- 2) **Directionality is explicit.** For several features (e.g., *Optic Glioma*, *Skeletal Dysplasia*) high values (red) systematically push the prediction toward one class, while low values (blue) move it toward the other. This gives domain experts not only a list of important features, but also the *polarity* of their effect.
- 3) **Heterogeneous features are context dependent.** Mid ranked variables such as *Inguinal Freckles*, *Axillary Freckles*, and *Dermal Neurofibromins* show a wider spread of SHAP values, meaning that the model sometimes uses them as supporting evidence and sometimes ignores them. This is typical for heterogeneous clinical cohorts and further supports the choice of a flexible model such as XGBoost.

Consequently, the SHAP analysis provides us to an instance aggregated explanation that complements the instance specific LIME view and the model level PFI view.

Figure 4 shows the LIME explanation generated for a single instance. The bars indicate the signed contribution of each feature to the predicted class for that specific patient record. In this example, *Hypertension* exerts the strongest positive effect, meaning that, conditional on the other attributes, its value pushes the XGBoost model toward the final decision. Conversely, a higher value in *Lisch Nodules* contributes negatively, reducing the model's confidence in the predicted class. Other clinical variables, such as *Scoliosis*, *Optic Glioma*, and *Age at First Diagnosis*, provide smaller but still positive support. This confirms that the model's output is the result of several moderately informative features combined in an additive way, rather than of a single

TABLE 4. Final classification results on the NF1 clinical dataset.

Model	Accuracy	Precision	Recall	F-score
XGBoost	0.804082	0.803572	0.802119	0.802170
RandomForest	0.710576	0.709310	0.704440	0.705285
LogisticRegression	0.555132	0.551859	0.551400	0.551192
LLM-Baseline	0.445946	0.456361	0.473139	0.403066
LLM-LoRA	0.543919	0.271959	0.500000	0.352298

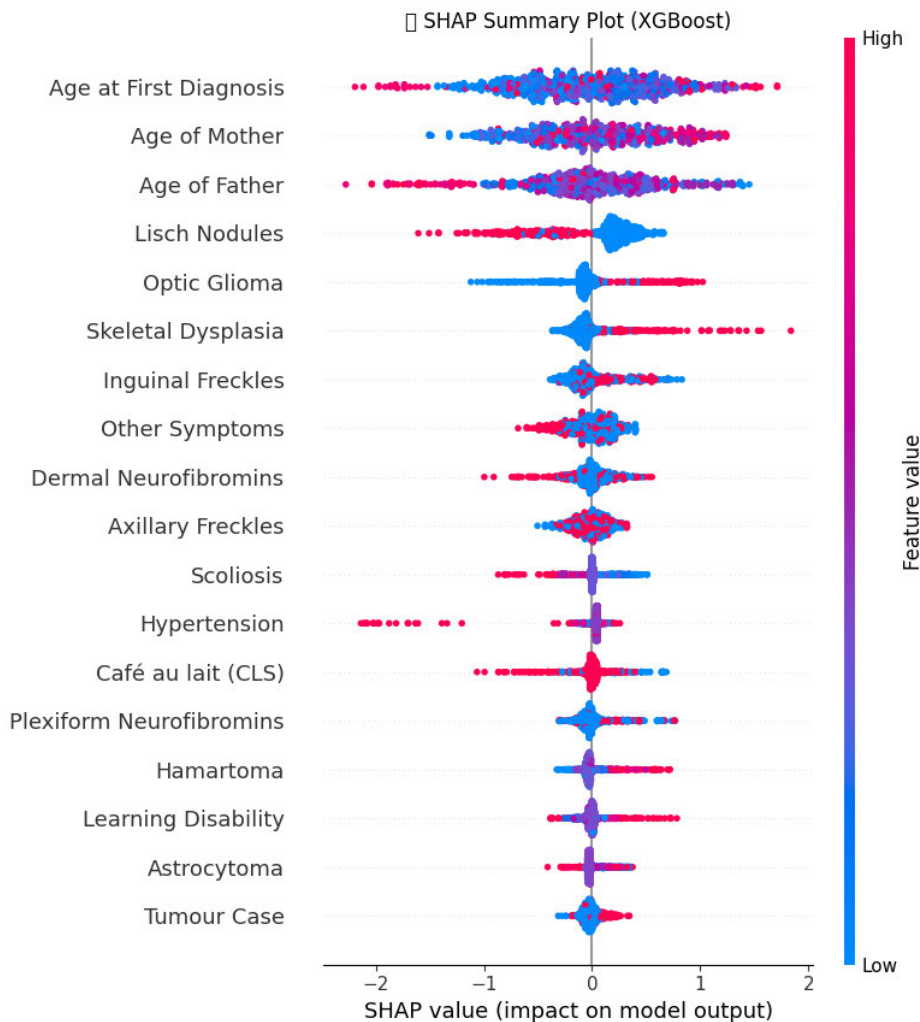


FIGURE 3. SHAP summary plot for the XGBoost model.(Features are ordered by overall impact; color indicates the original feature value (red = high, blue = low), and horizontal spread captures how much the feature shifts the prediction across the dataset.).

dominating predictor. LIME therefore offers *per-sample auditability*: a clinician or data steward can see precisely which feature values tipped the decision for this case. While LIME is inherently local, PFI provides a model agnostic, *global* view of feature relevance. In Fig. 5, features are ranked by the degradation in model performance when their values are randomly permuted. The top positions are occupied by *Age at First Diagnosis*, *Age of Mother*, and *Age of Father*, followed by *Lisch Nodules* and *Other Symptoms*. This ranking has two important implications.

First, it is clinically plausible that age related variables explain a substantial fraction of variation in NF1 presentation, and it is reassuring that the model recovers this pattern. Second, features at the bottom of the list (e.g., *Hamartoma*, *Learning Disability*) show negligible mean importance in this particular dataset. It means that they do not add discriminative power to the final classifier. XAI techniques are applied to the same model, the convergence of their output is important. Furthermore, age related variables and a subset of NF1 specific clinical signs (e.g.,

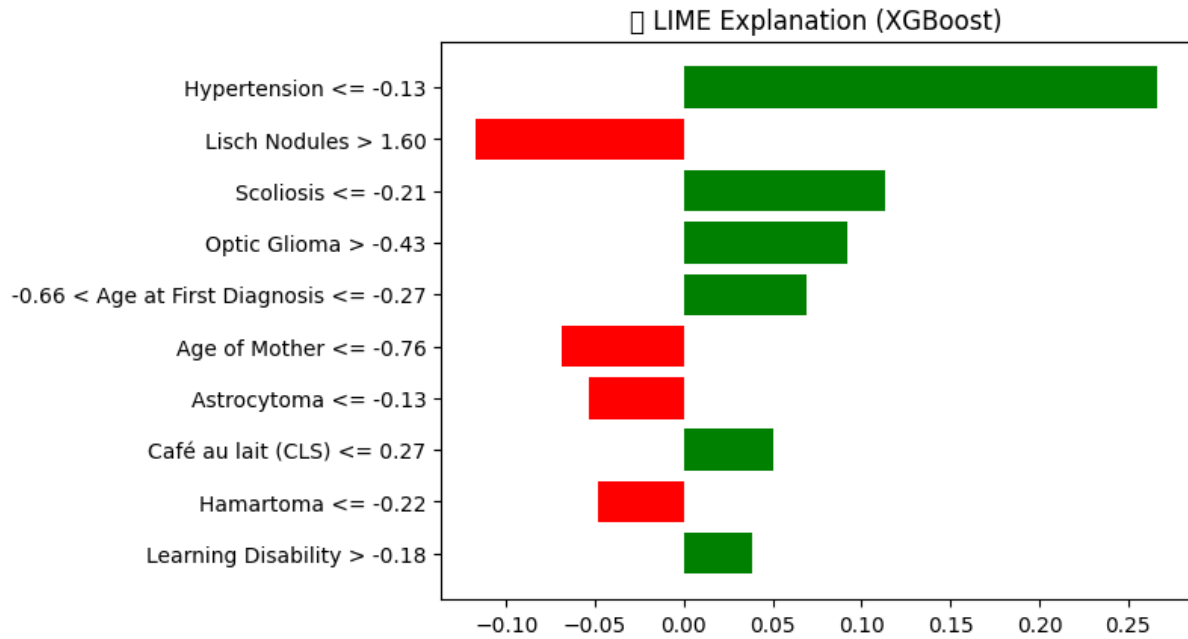


FIGURE 4. LIME based local explanation for the XGBoost model. (Green bars indicate features that increase the predicted score for the selected instance; red bars indicate features that decrease it).

Lisch nodules, optic glioma) repeatedly appear as influential. It can be inferred that the model is learning clinically plausible patterns rather than overfitting to synthetic noise introduced by SMOTE, Gaussian perturbation, or bootstrap resampling. It shows that the proposed framework is not only performant but also auditable, and that explanations can be generated at different granularity levels (local, global, and attributional) for downstream stakeholders.

F. LLM ASSISTED SYNTHESIS OF XAI OUTPUTS

In the final step of the proposed framework, the LLM-based narrative generator, specifically the microsoft/Phi-3-mini-4k-instruct model are employed. It automatically synthesize the heterogeneous XAI outputs (SHAP, LIME, and PFI) into a coherent, human readable explanation. This step takes as input the structured artefacts produced by the best model (XGBoost) and verbalizes them, correctly recognizing that “Age at First Diagnosis” and “Age of Mother” emerge as dominant factors across SHAP and LIME, and that their agreement strengthens the reliability of the learned decision logic (Figure 6). In this way, the LLM connects low level attribution scores with higher level model understanding. It prevents the user from having to combine numeric feature importances, local explanations, and data augmentation details by hand. It preserves the rigor of model agnostic XAI, but it packages the findings in a form that is immediately suitable for reporting, documentation, or clinician facing summaries. Thereby, It improves the interpretability and communicability of the overall system without altering its underlying predictions.

In detail, we adopt two scenarios as named initial and final setting to evaluate the results of LLM’s output. Initial results from using the model directly to generate an explanation without any prompt input. In the second scenario, some constraints are applied to the prompt, allowing for more optimized output. the output of the given scenarios is shown in table 5.

It is instantiated with two separate LLM generated narratives per instance, both produced by Phi-3-mini-4k-instruct. In the final setting, we explicitly constrained the prompt so that “each bullet MUST start with the feature name in quotes and MUST be followed by one short explanatory sentence describing its global effect (direction or relative importance). Bullet points that only list the name without any explanation are not allowed.” With using this prompt, a global narrative conditioned on SHAP, permutation importance summaries, and a local narrative conditioned on LIME outputs are generated for a single instance. Then it is assessed their quality using metrics as faithfulness, hallucination rate, and consistency at both feature and text levels. Table 6 shows that the revised prompting strategy leads to a substantial gain in both global and local faithfulness, and thus in overall explanation quality. The initial setting yields only moderate faithfulness scores (0.417–0.583 globally and 0.600–0.800 locally) because the model tends to focus on a subset of highly salient features and only partially covers the XAI derived feature sets. The final setting with two dedicated prompts and a bullet+sentence constraint consistently reaches 0.909 globally and 1.000 locally. It is indicated almost complete coverage of XAI identified features. In both settings the hallucination rate remains zero,

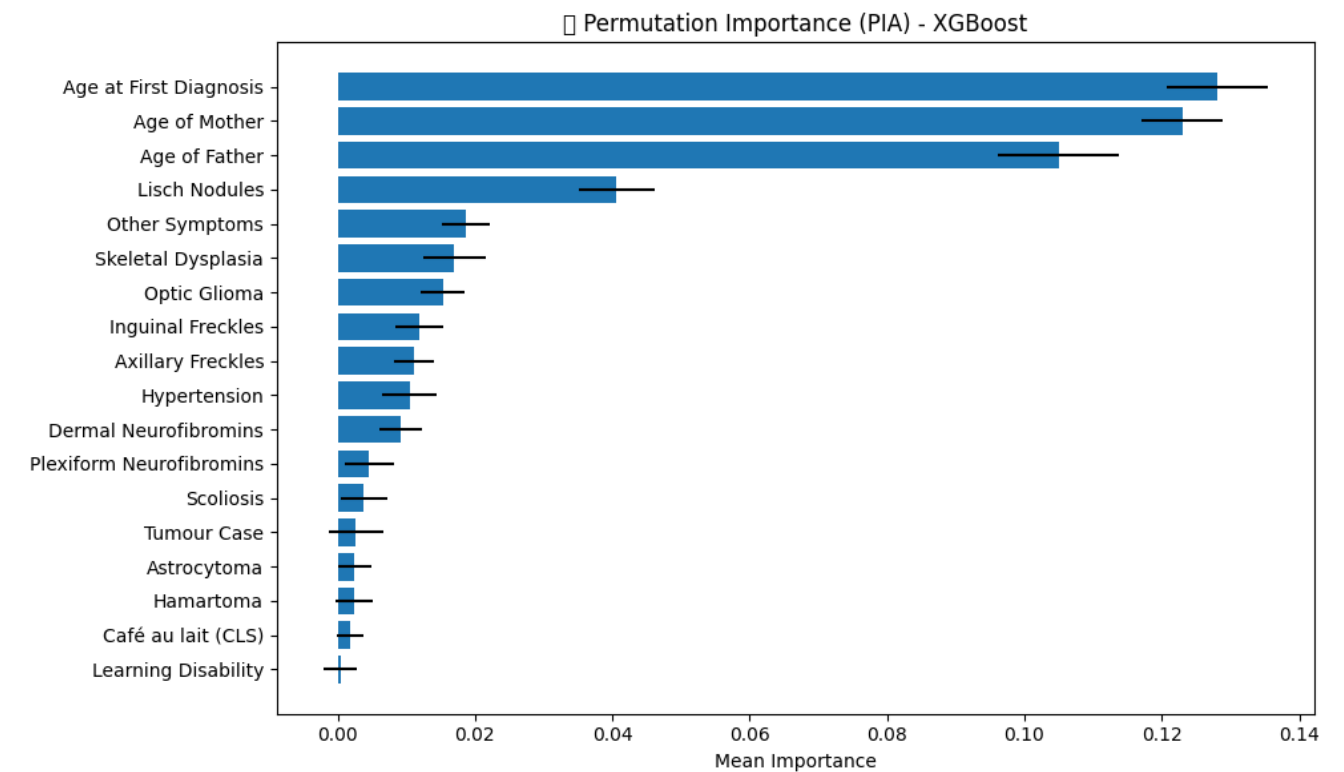


FIGURE 5. PFI for the XGBoost model. (Bars show the mean drop in performance when the corresponding feature is permuted; black lines indicate variability over repeats.)

TABLE 5. Qualitative comparison between the initial LLM prompting configuration and the final configuration.

Setting	Global narrative (qualitative)	Local narrative (qualitative)
Initial	The global behavior of the XGBoost model, as indicated by the SHAP values and permutation importance, reveals that 'Age at First Diagnosis' is the most influential feature overall, with a mean absolute SHAP value of 0.531976. This suggests that the age at which a patient first receives a diagnosis plays a critical role in predicting the 'Case Type'. Following closely, 'Age of Mother' and 'Age of Father' are also significant, with SHAP values of 0.422175 and 0.418548, respectively. These features underscore the potential genetic or familial factors contributing to the condition.	The local explanation for the specific instance, where the model predicted a 'Case Type' of 0, shows that the most influential features are 'Lisch Nodules' with a weight of 0.116320, 'Scoliosis' with 0.102325, and 'Age at First Diagnosis' with 0.099669. The negative weights associated with 'Tumour Case', 'Astrocytoma', and 'Hypertension' indicate that the presence of these conditions or symptoms would decrease the probability of the predicted 'Case Type'. Conversely, the positive weights for 'Café au lait (CLS)' and 'Skeletal Dysplasia' suggest that these features increase the likelihood of the predicted class.
Final	* "Age at First Diagnosis" * This feature has the highest mean absolute SHAP value, indicating it is the most influential factor in predicting the 'Case Type'. * "Age of Father" * With the second highest mean absolute SHAP value, this feature also plays a significant role in the model's predictions. * "Age of Mother" * Ranking third in terms of mean absolute SHAP value, this feature contributes to the model's age at First Diagnosis' predictive power. * "Axillary Freckles" * This feature has a notable mean importance in the permutation importance analysis, suggesting it is a key predictor. * "Dermal Neurofibromins" * With the highest mean importance in the permutation importance analysis, this feature is a critical predictor in the model. * "Inguinal Freckles" * This feature, while less influential than the top three, still holds a significant role in the model's predictions. * "Lisch Nodules" * Despite its lower mean importance, this feature is still a relevant predictor in the model. * "Optic Glioma" * This feature has a moderate mean importance, indicating its role in the model's predictions. * "Other Symptoms" * With a moderate mean importance, this feature contributes to the model's predictive power. * "Tumour Case" * This feature, while less influential than the top three, still holds a significant role in the model's predictions.	* "Age at First Diagnosis" has a negative weight of -0.192857, indicating that an earlier age at first diagnosis is associated with a lower probability of the predicted class (0). * "Age of Father" has a negative weight of -0.077521, suggesting that a younger father's age is associated with a lower probability of the predicted class (0). * "Age of Mother" has a negative weight of -0.092066, implying that a younger mother's age is associated with a lower probability of the predicted class (0). * "Axillary Freckles" has a negative weight of -0.039752, indicating that the presence of axillary freckles is associated with a lower probability of the predicted class (0). * "Dermal Neurofibromins" has a negative weight of 0.079203, suggesting that the presence of dermal neurofibromins is associated with a lower probability of the predicted class (0). * "Hypertension" has a positive weight of 0.229842, indicating that hypertension is associated with a higher probability of the predicted class (0). * "Learning Disability" has a positive weight of 0.043960, suggesting that a learning disability is associated with a higher probability of the predicted class (0). * "Lisch Nodules" has a positive weight of 0.108740, implying that the presence of Lisch nodules is associated with a higher probability of the predicted class (0). * "Optic Glioma" has a negative weight of 0.

but feature level consistency improves from 0.800 to 1.000. Furthermore, the number of mentioned features increases from 8–10 to 12–12. Textual consistency also stays high in both cases (0.907 vs. 0.893). It is suggested that these gains

in alignment and coverage are achieved without substantially altering the overall narrative style. Consequently, these results indicate that simple prompt engineering choices can greatly improve how well LLM generated narratives align with XAI

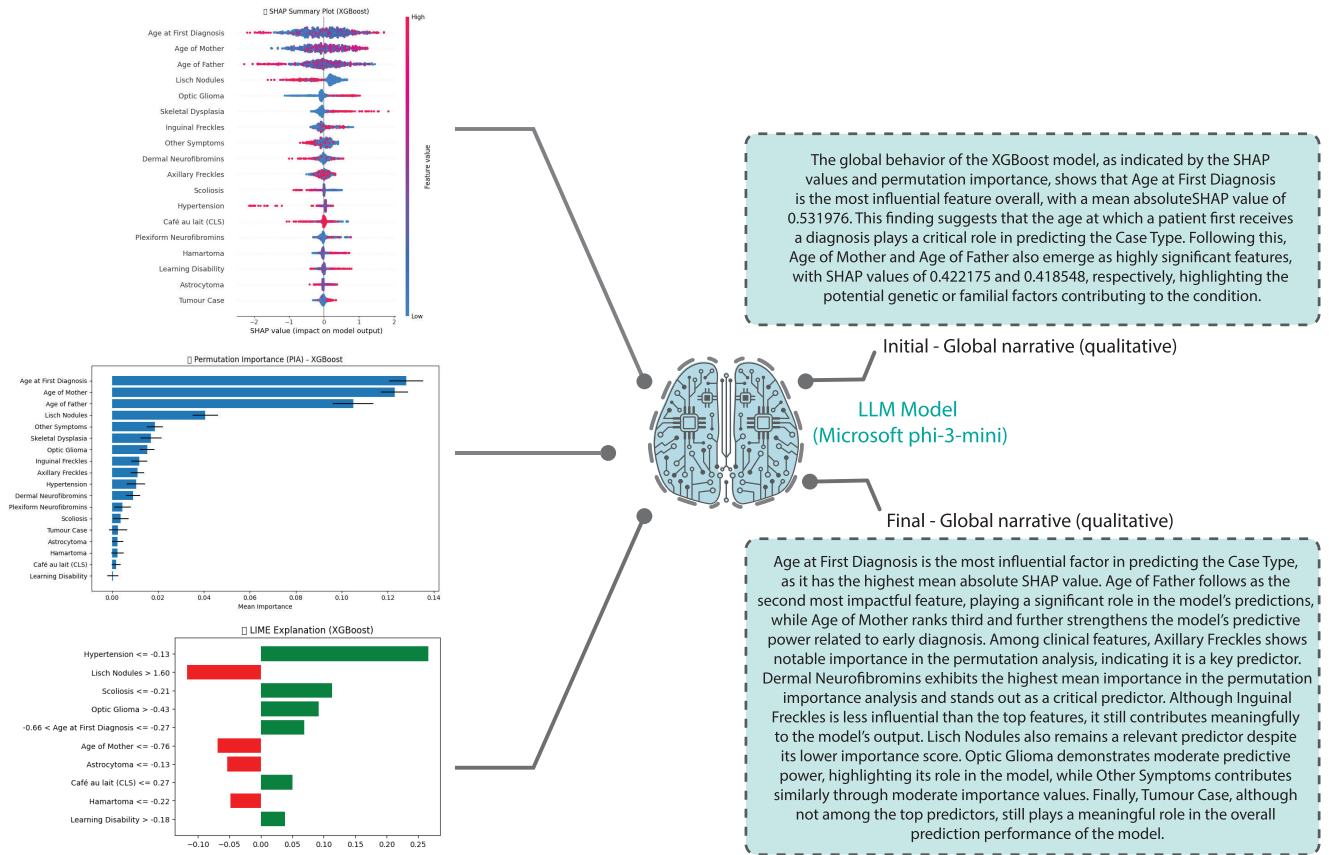


FIGURE 6. LLM outputs explanation of XAI models.

based model explanations. In particular, forcing the model to systematically cover XAI identified features increases this alignment while preserving the stability of the explanations.

V. DISCUSSION

The empirical results demonstrate that the proposed framework is effective for learning from a small and originally imbalanced NF1 clinical dataset. The proposed framework including data cleaning, correlation based feature reduction, class rebalancing with SMOTE, lightweight data augmentation, stratified 5 fold validation, and model level explainability collectively contributed to higher and more stable performance. Experimental results shows that XGBoost classifier achieved an accuracy of 0.804 and a F - score of 0.802, outperforming all other models. The comparison across classical models further supports this conclusion. Random Forest, which benefits from ensembling but does not iteratively reweight hard examples, reached an accuracy of 0.711 and a F - score of 0.705 respectable, but clearly remain below XGBoost. Logistic Regression lagged behind both tree based approaches (accuracy 0.555, F_1 0.551). It is indicated that the decision surface in this clinical feature space is not predominantly linear. This ranking is consistent with the hypothesis that the proposed framework exposes richer, non linear patterns once the data are balanced and augmented,

and that boosting is better positioned than bagging or linear models to exploit those patterns.

The LLM based prediction models remains under the classical ML algorithms. The LLM baseline obtained only 0.446 accuracy and 0.403 F_1 , showing that, without task specific conditioning, language models do not automatically acquire the inductive bias needed for structured clinical classification. The LoRA adapted LLM improved in terms of recall (0.500) and overall accuracy (0.544), but its precision dropped markedly (0.272), leading to a low F_1 of 0.352. This pattern suggests that parameter efficient fine tuning helps the LLM become more sensitive to the positive class. But It does not fully align its output distribution with the true class distribution. This limitation stems primarily from the mismatch between model architecture and data availability. Fine-tuning a large language model on a small clinical dataset of only 241 samples is insufficient to bridge the fundamental gap between text-based pretraining objectives and the structured numerical reasoning required for tabular classification. Consequently, for this dataset, it is inferred that classical tree based methods remain the most reliable option.

The performance obtained in this study can be compared with the results reported by Sharafi [3] who evaluated several ML methods on the same NF1 clinical symptom dataset. In their work, the XGBoost classifier reached an

TABLE 6. Quantitative comparison of automatic LLM explanation metrics between the initial and final prompting configurations.

Initial setting (single global+local prompt)	Final setting (separate global/local prompts + bullet+sentence constraint)
Global faithfulness (Narr. 1 / Narr. 2)	Global faithfulness (Narr. 1 / Narr. 2)
0.417 / 0.583	0.909 / 0.909
Local faithfulness (Narr. 1 / Narr. 2)	Local faithfulness (Narr. 1 / Narr. 2)
0.600 / 0.800	1.000 / 1.000
Hallucination rate (Narr. 1 / Narr. 2)	Hallucination rate (Narr. 1 / Narr. 2)
0.000 / 0.000	0.000 / 0.000
Feature level consistency (Jaccard)	Feature level consistency (Jaccard)
0.800	1.000
Textual consistency (cosine similarity)	Textual consistency (cosine similarity)
0.907	0.893
#Features mentioned (Narr. 1 / Narr. 2)	#Features mentioned (Narr. 1 / Narr. 2)
8 / 10	12 / 12

accuracy of approximately 62.86%. In proposed pipeline, the XGBoost model achieved an accuracy of 80.41% and a F_1 of 80.22%, which represents a substantial improvement over the previously reported figures. Thus, the comparison indicates that proposed frameworks improves the prediction performance itself for this clinical prediction task.

By running SHAP, PFI, and LIME on the *best* model, it verifies that the gain in accuracy is not achieved at the expense of interpretability. All three XAI views identified age related variables (e.g., *Age at First Diagnosis*, *Age of Mother*) and selected NF1 symptoms as major drivers of the model's decisions. This agreement across methods increases trust that the improvements observed after SMOTE and augmentation. The LLM based narrative step (Phi-3-mini-4k-instruct) enables to add a explanation of this XAI outputs. In this step, it integrates numeric attributions, local explanations, and the data preparation rationale into a single textual account. This step gives valuable information to non ML stakeholders because it turns a technically sound pipeline into one that is also communicable. With development of the LLM's, LLM integrated works emerge to literature. The XAI4LLM [52] framework use LLM as predictor and compare performance to ML models. Different from the XAI4LLM [52], the proposed framework treats the LLM as a narrative layer on top of SHAP, LIME, and PFI based explanations and explicitly quantifies explanation quality via global and local faithfulness, hallucination rate, and feature-text consistency metrics. Furthermore, the proposed framework numerically assess the alignment between its generated narratives and the underlying XAI evidence. TT-XAI [53] is designed for long, unstructured electronic health records and uses keyword distillation. The authors use a BERT classifier, focus LIME, and a large LLM to generate clinical explanations [53]. In contrast, the T-XAI, the proposed framework operates on structured clinical risk factors and first computes SHAP, LIME, and PFI feature attributions, then asks a smaller LLM to transform these attributions into a concise and coherent narrative. Whereas TT-XAI evaluates explanations mainly with deletion based fidelity curves, LLM self-scores, and expert ratings. The proposed framework additionally report global and local faithfulness scores, hallucination rate, and

feature text overlap metrics, giving more direct control over how closely the generated text follows the underlying XAI evidence.

Overall, the discussion can be summarized in three points. First, the proposed stepwise framework directly improves model quality: balancing + augmentation made it possible for XGBoost to reach the 0.80 F_1 range, which would have been unlikely on the raw, imbalanced data. Second, the framework makes model comparison meaningful: because every model is trained on the same processed data and evaluated with stratified 5 fold CV, the superiority of boosted trees over bagging, linear models, and LLMs is attributable to modeling capacity rather than to data leakage or evaluation bias. Third, the framework preserves transparency: the combination of XAI tools and an LLM based summarizer produces explanations at different levels of abstraction. It ensure that model's higher accuracy converts to explainability.

Despite the promising findings, this study has several limitations that should be acknowledged. The proposed framework focuses exclusively on tabular clinical features and does not incorporate imaging, genomic, or longitudinal data, which could provide deeper diagnostic insights for NF1. Another limitation is the LLM based explanation layer that relies on a compact instruction tuned model. This layer's interpretive output has not been systematically validated with clinicians, which limits conclusions about its applicability in real world decision support. Nevertheless, the clinical utility of the narrative layer is supported quantitatively: the final prompting configuration achieves a global faithfulness of 0.909, local faithfulness of 1.000, a zero hallucination rate, and a feature-level consistency of 1.000 (Table 6), confirming that the generated narratives accurately reflect the underlying XAI evidence without introducing unsupported clinical claims. A formal clinician evaluation study is planned as part of future work to empirically validate the practical benefit of this layer in real clinical decision-making settings.

VI. CONCLUSION

This study introduce a structured, end to end framework for classifying familial and sporadic NF1 cases from a small and initially imbalanced clinical dataset. The proposed

framework combined several steps such as data cleaning, preprocessing, class balancing and augmentation, model definition, stratified 5 fold cross validation, and multi view explainability. This framework is applied to health domain with using standard tabular data. In experiments, the XGBoost model achieved the best overall performance (accuracy ≈ 0.80 , $F - score \approx 0.80$), clearly outperforming Random Forest and Logistic Regression and markedly improving on previously reported results for the same dataset. The LLM baseline and its LoRA adapted variant both underperformed the tree based models. This results shows that, for NF1 type clinical symptoms, proposed framework offers a promising framework for decision makers in health domain. The LLM component proves useful in a different role turning heterogeneous XAI artefacts (SHAP, LIME, PFI) into coherent, human readable narratives. In this sense, the LLM acted as an interpretability amplifier rather than as a primary predictor, helping to make the final model suitable for documentation and potential clinician facing use.

In future work, It can be extended this framework to multi center NF1 data, test additional parameter efficient finetuning schemes for LLMs, and explore semi supervised augmentation to further reduce the dependence on manually collected clinical cases.

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